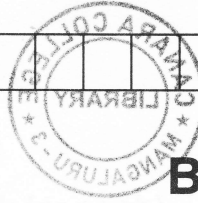


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BCACACS 201

**Second Semester B.C.A. Degree Examination, June/July 2026
(SEP) (2024 – 2025 Batch Onwards)
DATA STRUCTURES**

Time : 3 Hours

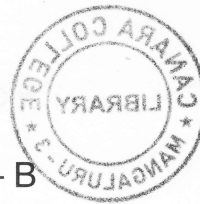
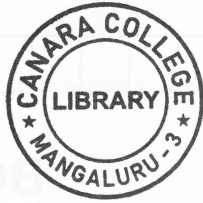
Max. Marks : 80

Instruction : Answer **any ten** questions from Part – **A** and **any one** full question from **each** Unit in Part – **B**.

PART – A

1. a) Define primitive data structure. Give an example. (10×2=20)
 - b) Consider the linear array LA(1:10) and find the number of elements in the array.
 - c) Define Recursion.
 - d) Write one advantage and disadvantage of binary search.
 - e) What do you mean by dynamic memory allocation ? Write any 2 functions used in 'C' for dynamic memory allocation.
 - f) Write the advantages of linked list over arrays.
 - g) How does Stack differ from Queue ?
 - h) Define i) infix notation, ii) prefix notation.
 - i) Define dequeue. What are its types ?
 - j) Define binary tree. Give an example.
 - k) What is Graph ? Give an example.
 - l) Define adjacency matrix and path matrix.

P.T.O.

**Unit – I**

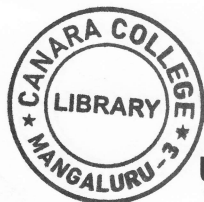
2. a) Explain sparse matrix with example. (4+5+6)
 b) Write an algorithm for TOWER OF HANOI using recursion with example.
 c) Explain selection and iteration logic in detail with example.
3. a) Write an algorithm for insertion sort. (4+5+6)
 b) Trace the following list of numbers using Selection sort technique :
 77, 33, 44, 11, 88, 22, 66, 55.
 c) Explain the representation of two-dimensional array in memory.

Unit – II

4. a) Write an algorithm for binary search. (5+5+5)
 b) Explain the memory representation of linked list.
 c) Explain Circular linked list.
5. a) What is linked list ? Explain its different types. (5+5+5)
 b) Write an algorithm to insert an item after a given node of a linked list.
 c) What is linear search ? Write the algorithm for linear search.

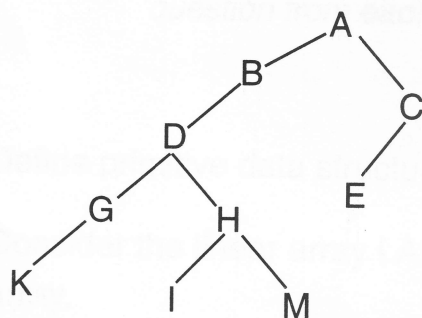
Unit – III

6. a) Write an algorithm to PUSH and POP operation onto a stack using arrays. (4+5+6)
 b) Write a note on i) Circular queue, ii) Deque.
 c) Convert the following infix expression into postfix expression using stack status.
 $Q : A + (B * C - (D / E ^ F) * G (* H)).$
7. a) Convert the given infix expression to postfix form (4+5+6)
 i) $A * B + (C ^ D / E / F)$
 ii) $(A + B) * (C - D) / E * F.$
 b) Write an algorithm to perform insertion operation on circular queue using array.
 c) Evaluate the following postfix expression showing the stack status.
 $P : 3, 5, +, 6, 4, -, *, 4, 1, -, 2, ^, +.$



Unit – IV

8. a) Define the terms with respect to tree. (1 mark for **each**) (5+5+5)
i) leaf node ii) depth of a tree iii) descendent iv) siblings v) path.
- b) Draw the binary search tree for the following data and traverse in all 3 methods
14, 15, 4, 9, 7, 18, 40, 35, 16, 13.
- c) Write an algorithm for depth first search (DFS) for a graph.
9. a) Write the preorder, inorder and postorder traversal for the given tree. (5+5+5)



- b) Explain Linked Representation of a graph with example.
- c) Write an algorithm for Breadth First Search for Graph.
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